

1 Preface

From today(Nov 11) i will begin to re-learn quantum mechanics and start my note working ,try to fix the whole quantum world in my head.

Today is Dec 17 and I almost know the 7,8,chapter. Now I'd like to list some of important points in QM From Chapter 1 to Chapter 8:

Chapter 1:

I : Schrödinger equation : $-\frac{\hbar^2}{2m}\nabla^2\psi + V\psi = \hat{H}\psi$

II : propagator : $\psi(t) = \sum_n C_n \phi_n e^{-iE_n t/\hbar}$ we will redefine the C in the behind

III : The definition of **Stationary state(3.3)**

Chapter 2:

I : The property of particle in one dimensional potential field

II : The parity of the wavefunction under a symmetry potential in space is even or odd

III : The transmission and reflection is important define

IV : The important calculation of potential well(both the square delta and oscillator)

V : The energy of bound state

2 Wavefunction

2.1 Begin of the probability wave

Describe the probability of a particle that will appear in a place. And from the principle of quantum mechanics, consider a situation that doesn't produce any or annihilate any particle.

$$\int \phi^* \phi \quad \varepsilon = 1 \quad (1)$$

That means the whole probability of a particle is 1 for every possible position. Here are some equations that will be used in the next learning:

$$\overline{(\Delta u)^2} = \overline{(u^2)} - (\bar{u})^2 \quad (2)$$

$$\overline{\Delta u} = \sqrt{\overline{u^2} - \bar{u}^2} \quad (3)$$

These are the simplest math equations but will be used in **uncertainty relation**.

2.2 wave function

we call the wave function which is differ in a constant multiplied are the same function:

$$\phi = C\phi \quad (4)$$

so we call the principle is **the probability of the function waves are relatively**.

2.3 momentum

We all know how to define the momentum of a particle , but (consider it is the relearn note) we know the uncertainty principle,so by using **Fourier transformation**:

$$f(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} f(x)e^{-ikx} dx \quad (5)$$

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} f(k)e^{ikx} dk \quad (6)$$

and define the momentum:

$$\phi(p) = \frac{1}{\sqrt{2\pi\hbar}} \int \phi(x)e^{\frac{-ipx}{\hbar}} dx \quad (7)$$

What do $\phi(p)$ express?It express the **format of wave function in the space of momentum**.So we can calculate the momentum by:

$$\bar{p} = \int_p \phi(p)^* p \phi(p) \quad \epsilon \quad (8)$$

This integerallarge is defined on the space of momentum,the reason of why we not to choose to calculate the average momentum by using : $\int \phi(x)^* p \phi(x)$ is the momentum can not be certain with x at the same time.

And we also consider about the de Broglie relation:

$$p = \frac{h}{\lambda} \quad (9)$$

2.4 uncertainty relation

The strict define of uncertainty relation:

$$\Delta A \Delta B \geq \frac{1}{2} |\overline{[A, B]}| \quad (10)$$

The $\Delta A \Delta B$ here are the $\sqrt{(\Delta A)^2 \times (\Delta B)^2}$.

2.5 Average of Mechanic Variable

The define is :

$$\int \phi^* A \phi = \overline{A} \quad (11)$$

2.6 coordinate representation

How to express coordinate representation? It just like the relationship between physics object and coordinate base. Like how to get the part of a tensor or a part of a vector. The $\phi(\mathbf{t})$ is the physics object and $\phi(\mathbf{x})$ is this object in x space, and $\phi(\mathbf{p})$ is the object in momentum space.

And the base of these projection are the base in Hilbert space.

Hilbert space: the every possible state of wave function ϕ .

CSCO: a set of commuting operator, and the co-eigenstate is a base of Hilbert space.

3 Shrödinger equation

$$i\hbar \frac{\partial \phi(r, t)}{\partial t} = [-\frac{\hbar^2}{2m} \nabla + V(r, t)] \phi(r, t) \quad (12)$$

3.1 propagator

Because Shrödinger equation only have the 1 class derivative of time t so if we set the wave function of **free particle** at $t = 0$: $\phi(r, 0)$ we can get we any time wave function $\phi(r, t)$:

$$\phi(r, t) = \frac{1}{2\pi\hbar^{\frac{3}{2}}} \int \phi(p) e^{\frac{ipr - iEt}{\hbar}} d^3p \quad E = \frac{p^2}{2m} \quad (13)$$

$$\phi(r, 0) = \frac{1}{2\pi\hbar^{\frac{3}{2}}} \int \phi(p) e^{\frac{ipr}{\hbar}} d^3p \quad (14)$$

$$\text{reverse tranformation: } \phi(p) = \frac{1}{2\pi\hbar^{\frac{3}{2}}} \int \phi(r, 0) e^{\frac{-ipr}{\hbar}} d^3r \text{ put back to (13)} \quad (15)$$

$$\phi(r, t) = \frac{1}{2\pi\hbar^3} \int d^3p \int d^3r' \phi(r', 0) e^{\frac{ip(r-r') - iEt}{\hbar}} \quad (16)$$

The reason of from eq(14) to eq(15) is **Fourier transformation don't rely on time t** and for any common situation:

$$\psi(r, t) = \frac{1}{2\pi\hbar^3} \int d^3r' \int d^3p \phi(r', t') e^{\frac{ip(r-r') - iE(t-t')}{\hbar}} \quad (17)$$

rewrite it into form :

$$\psi(r, t) = \int d^3r' G(r, r', t, t') \phi(r', t') \quad (18)$$

$$G = \frac{1}{2\pi\hbar^3} \int e^{\frac{ip(r-r') - iE(t-t')}{\hbar}} d^3p \quad (19)$$

and the G in there is called propagator. For an eigenstate of $\hat{H}$ the wave function can be write as: $\psi(r, t) = \frac{1}{2\pi\hbar^3} e^{\frac{ipr - iEt}{\hbar}}$.

Now look back at eq(18) we can express this equation as : Wave function of time at t is all wave function of the particle in time t' and r' coherent superposition then spreaded to r .

3.2 Time Factor

We set $\psi(r, t) = \psi(r)f(t)$ and consider a simple situation : V in $\hat{H}$ don't explicit t. And $\psi(r, t)$ is the eigenstate of E.

$$i\hbar\psi(r)\frac{\partial f(t)}{\partial t} = [-\frac{\hbar^2}{2m}\nabla + V]\psi(r, t) \quad (20)$$

$$Ef(r, t) = [-\frac{\hbar^2}{2m}\nabla + V]\psi(r, t) \quad (21)$$

$$f(t) = e^{\frac{-iEt}{\hbar}} \quad (22)$$

so the $\psi(r, t)$ can be rewrite as : $\psi_E(r)f(t)$ For a more normalized situation, we can write any $\psi(r, t)$ into eigenstate of E: $\psi_{En}(r)$:

$$\psi(r, t) = \sum_n C_n \psi_{En} f_n(t) = \sum_n C_n \psi_{En} e^{\frac{-iE_n t}{\hbar}} \quad (23)$$

$$|C_n|^2 = \int \psi_{En}^* \psi(r, 0) \quad (24)$$

3.3 Stationary state

We call a wave function in form $\psi(r, t) = \psi(r)e^{-i\frac{Et}{\hbar}}$ as the stationary state wave function. When a particle stay in stationary state the energy of this particle is exactly can be determined. And the particle have these nature satisfied when it is a stationary particle:

(1) the probability density is not go with time:

$$\psi^* \psi = \rho(r) \quad (25)$$

$$(26)$$

(2) the probability flux is not go with time :

$$j_n(r, t) = \frac{i\hbar}{2m} [\psi_n \nabla \psi_n^* - \psi_n^* \nabla \psi_n] = j_n(r) \quad (27)$$

(3)any average of dynamical variables which don't dependent on time will not dependent on time:

$$\overline{F} = \int \psi(r, t)^* F \psi(r, t) = \int \psi(r)^* F \psi(r) \quad (28)$$

(4)any observed value from observable dynamical variables don't explicit t will not dependent on time.

4 Operator

4.1 The Operator Algebra

Operators normally don't satisfy the law of multiple exchange. So we define the commutator:

$$[A, B] = AB - BA \quad (29)$$

And also can be defined the linearity of operators:

$$(c_1 A + c_2 B)\psi = c_1 A\psi + c_2 B\psi \quad (30)$$

4.2 commutator

Commutator is one of the important concept in QM, and there are some qualities commutator have:

$$[x, y(p(x))] = i\hbar \frac{\partial y}{\partial p(x)} \quad (31)$$

$$[p(x), y(x)] = -i\hbar \frac{\partial y}{\partial x} \quad (32)$$

The equation(31) is the equation in coordinate representation p, and equation(32) is in the coordinate representation x. So we can find the basic law:

$$[A(x), B(A(x))] = i\hbar \frac{\partial B}{\partial A(x)} \quad (33)$$

and from the upper discussion we know there is a strong relationship between commutator and uncertainty relation at equation(10).

And the function of a variable will commuted with another function of the same variable

$$[A(f), B(f)] = 0 \quad (34)$$

4.3 Some unique transform in QM

Most of mechanics operators is hermited, $\hat{A}^\dagger = \hat{A}$ for define

$$\int \phi^* \hat{A}^\dagger \psi = \int (\hat{A} \psi)^* \phi = \int \psi^* \hat{A} \phi \quad (35)$$

4.4 angular momentum in QM

The operator of angular momentum: $\hat{l} = r \times \hat{p} = r^i \hat{p}^j \varepsilon_{ijk}$ Rewrite them in a sphere coordinate system, noticed that the form of l_z is simple and **we know now, the eigenstate of l_z is a Spherical harmonic function Y_{lm} . NOTICE** : the eigentstate of l_z can be a part of Sphereical harmonic function : $\psi = \frac{1}{\sqrt{2\pi}} e^{im\phi}$, and the eigenstate of l^2 can be another ppart of Spherical harmonic function. **So Spherical harmonic function is the co-eigenstate of both opeator : $\hat{l}_z, \hat{l}^2$.**

$$\hat{l}_z Y_{lm} = m\hbar Y_{lm} \quad (36)$$

$$\hat{l}^2 Y_{lm} = l(l+1)\hbar^2 Y_{lm} \quad (37)$$

$$[\hat{l}^2, \hat{l}_i] = 0 \quad (38)$$

$$[l_i, x^j] = i\hbar \varepsilon_{ijk} x^k \quad (39)$$

$$[\hat{l}_i, \hat{l}_j] = i\hbar \varepsilon_{ijk} \hat{l}_k \quad (40)$$

$$l_{\pm} Y_{lm} = \sqrt{l(l+1) + m(m \pm 1)} \hbar Y_{lm} \quad (41)$$

And the define of Shperiacl harmonic function:

$$Y_{lm}(\theta, \phi) = (-1)^m \sqrt{\frac{(l-m)!(2l+1)}{4\pi(l+m)!}} P_l^m(\cos\theta) e^{im\phi} \quad (42)$$

$$l = 0, 1, 2, \dots, m = -l, -l+1, \dots, l-1, l.$$

And the Orthogonal normalization condition is written as:

$$\int Y_{lm}^* Y_{l'm'} = \delta_{ll'} \delta_{mm'} \quad (43)$$

Now I will list a important example :

$$\hat{l}^2 - \hat{l}_z^2 = \hat{l}_x^2 + \hat{l}_y^2 \text{ and consider a situation } (\psi = Y_{lm}): \int \psi^* \hat{l}_x^2 \psi$$

$$= \int \phi^* \frac{1}{i\hbar} [\hat{l}_y, \hat{l}_z] \hat{l}_x \psi = \frac{1}{i\hbar} \int \psi^* (\hat{l}_y \hat{l}_z - \hat{l}_z \hat{l}_y) \hat{l}_x \psi$$

$$= \frac{1}{i\hbar} \int \phi^* \hat{l}_y \hat{l}_z \hat{l}_x \psi - \phi^* \hat{l}_z \hat{l}_y \hat{l}_x \psi$$

$$\text{consider } \hat{l}_z \hat{l}_x = \hat{l}_x \hat{l}_z + i\hbar \hat{l}_y$$

$$= \frac{1}{i\hbar} \int \psi^* \hat{l}_y (\hat{l}_x \hat{l}_z + i\hbar \hat{l}_y) \psi - \psi^* \hat{l}_z \hat{l}_y \hat{l}_x \psi$$

$$= \frac{1}{i\hbar} \int \psi^* \hat{l}_y \hat{l}_x \hat{l}_z \psi + i\hbar \psi^* \hat{l}_y^2 \psi - \psi^* \hat{l}_z \hat{l}_y \hat{l}_x \psi$$

$$= \frac{1}{i\hbar} \int \psi^* \hat{l}_y \hat{l}_x m\hbar \psi + i\hbar \psi^* \hat{l}_y^2 \psi - m\hbar \psi^* \hat{l}_y \hat{l}_x \psi$$

$$= \int \psi^* \hat{l}_y^2 \psi$$

so we get

$$\overline{\hat{l}_x^2} = \overline{\hat{l}_y^2} \quad (44)$$

Now consider the first equation :

$$\overline{\hat{l}_z^2} + \overline{\hat{l}_x^2} = 2\overline{\hat{l}_y^2} = 2\overline{\hat{l}_x^2}$$

$$\text{and } \int \psi^* \hat{l}^2 \psi = \int \psi^* l(l+1)\hbar^2 \psi = l(l+1)\hbar^2 ; \int \psi^* \hat{l}_z^2 \psi = \int (\hat{l}_z \psi)^* \hat{l}_z \psi = m^2 \hbar^2$$

so

$$\overline{\hat{l}_x^2} = \overline{\hat{l}_y^2} = \frac{1}{2}(l(l+1) - m^2)\hbar^2 \quad (45)$$

We know the average of angular momentum $\hat{l}_x$ and $\hat{l}_y$ is : $\int Y_{lm}^* \hat{l}_x Y_{lm} \hat{l}_x = \int Y_{lm}^* \frac{1}{i\hbar} [\hat{l}_y, \hat{l}_z] Y_{lm} = \frac{1}{i\hbar} \int Y_{lm}^* \hat{l}_y \hat{l}_z Y_{lm} - Y_{lm}^* \hat{l}_z \hat{l}_y Y_{lm} = 0$ so we get :

$$\overline{\hat{l}_x} = \overline{\hat{l}_y} = 0 \quad (46)$$

so we know the uncertainty relation between $\hat{l}_x$ and $\hat{l}_y$

$$\Delta \hat{l}_x \Delta \hat{l}_y = \sqrt{\overline{\hat{l}_x^2} - \overline{\hat{l}_x}^2} \sqrt{\overline{\hat{l}_y^2} - \overline{\hat{l}_y}^2} = \frac{1}{2}(l(l+1) - m^2)\hbar \quad (47)$$

4.5 Function of operator

We can define the function of operators as :

$$F(\hat{A}) = \sum_n \frac{F^n(0)}{n!} \hat{A}^n \quad (48)$$

such as : $e^{a \frac{d}{dx}}$ is a typicaly function of operator. And the effect of this function is :

$$e^{a \frac{d}{dx}} \psi(x) = \sum_n \frac{a^n}{n!} \frac{d^n}{dx^n} \psi(x) = \psi(x+a) \quad (49)$$

so $e^{a \frac{d}{dx}}$ is the operator that move the function point from x to x+a.

5 symmetry of evolutioning Dynamical variables

5.1 Conserved observable

If a average value of observable is not changed with time , thus this observable is a conserved observable.

$$\frac{d}{dt} \bar{A}(t) = \frac{1}{i\hbar} \overline{[A, H]} \quad (50)$$

So if the operator of this observable is comuted with Hamilton operator , this observable is a conserved observable. **Notice the detail of define described :AVERAGE VALUE is not go with time,not the classic type "value not go with time".The reason is the principle in QM is very different with classic mechanic.** And ofcourse a Conserved Observable can be a not certainll value,so the state of the operator observed can be the non-eigenstate of this operator. The key word is "average value".

5.2 Relationship Between Conserved Observable And Energy Level Degenerate

There are three Theorems:

(A) Set two non-commutated conserved observable F and G in a system, that means: $[F, H] = 0$; $[G, H] = 0$ but $[F, G] \neq 0$, thus the energy level usually degenerated.

(B) Inference from (A): If a conserved observable F in a system and there is a level of energy not degenerated, then ψ_E is the eigenstate of F.

(C) virial theorem:

$$2\overline{T} = \overline{\mathbf{r} \cdot \nabla V} \quad (51)$$

5.3 * Schrödinger picture and Heisenberg picture

The average value derivative with t is $\frac{d}{dt} \overline{A(t)} = \frac{1}{i\hbar} \overline{[A, H]}$, now we rewrite $\psi(r, t)$ as $\hat{U}(t, 0)\psi(r, 0)$ and the $\hat{U}(t, 0)$ here is the time shift operator and $\hat{U}(0, 0) = 1$. Now we put it back into Schrödinger equation:

$$\begin{aligned} i\hbar \frac{\partial}{\partial t} \psi(r, t) &= \hat{H} \psi(r, t) \\ i\hbar \frac{\partial}{\partial t} \hat{U}(t, 0) \psi(r) &= \hat{H} \hat{U}(t, 0) \psi(r) \\ i\hbar \frac{\partial}{\partial t} \hat{U}(t, 0) &= \hat{H} \hat{U}(t, 0) \end{aligned}$$

so we get (if $\hat{H}$ is not go with time) :

$$\hat{U}(t, 0) = e^{\frac{-i\hat{H}t}{\hbar}} \quad (52)$$

or ($\hat{H}$ go with time):

$$\hat{U}(t, 0) = e^{\frac{-i \int \hat{H} dt}{\hbar}} \quad (53)$$

and we can proof:

$$\begin{aligned} \int \phi(r, t)^* \psi(r, t) &= \int \hat{U}(t, 0)^* \phi(r, 0)^* \hat{U}(t, 0) \psi(r, 0) \\ &= \int \phi(r, 0)^* \phi(r, 0) = 1 \end{aligned} \quad (54)$$

For a more common situation we can say the **wave function is not go with time but operators are go with time**, and we rewrite $\phi(r, t)$ into $\hat{U}(t, 0)\psi(0)$. And the operator can be written in:

$$\begin{aligned} \overline{\hat{A}(t)} &= \int \hat{U}(t, 0)^* \psi(0)^* \hat{A} \hat{U}(t, 0) \psi(0) \\ &= \int \psi(0)^* \hat{U}(t, 0)^\dagger \hat{A} \hat{U}(t, 0) \psi(0) \\ &= \int \psi(0)^* \hat{A}(t) \psi(0) \end{aligned}$$

so we get the operator $\hat{A}(t)$ defined as

$$\hat{A}(t) = \hat{U}(t, 0)^\dagger \hat{A} \hat{U}(t, 0) \quad (55)$$

So the evolution equation of operators evoluto with time :

$$\begin{aligned} \frac{d}{dt} \hat{A}(t) &= \frac{d}{dt} (\hat{U}(t, 0)^\dagger \hat{A} \hat{U}(t, 0)) = \frac{d\hat{U}(t, 0)^\dagger}{dt} \hat{A} \hat{U}(t, 0) + \hat{U}(t, 0)^\dagger \hat{A} \frac{d\hat{U}(t, 0)}{dt} \\ &= \frac{1}{-i\hbar} \hat{U}(t, 0)^\dagger \hat{H} \hat{A} \hat{U}(t, 0) + \frac{1}{i\hbar} \hat{U}(t, 0)^\dagger \hat{A} \hat{H} \hat{U}(t, 0) = \frac{1}{i\hbar} \hat{U}(t, 0)^\dagger [\hat{A}, \hat{H}] \hat{U}(t, 0) \end{aligned}$$

Because $\hat{U}(t, 0)^\dagger \hat{H} \hat{U}(t, 0) = \hat{H}$ (easily proved with: $\hat{A}$ and function of $\hat{A}$ are commutated) so :

$$\frac{d}{dt} \hat{A}(t) = \frac{1}{i\hbar} [\hat{A}(t), \hat{H}] \quad (56)$$

Noticed that this is quite differernt from eq(49) this equation is not about average value but the equation of operator. And this kind of theory is called **Heisenberg Represent**. The equation : is called Heisenberg equation. As the evolution equation of operators.

5.4 The Relationship Between Conserved Observable and Symmetry

A.E.Noether: "every symmetry in nature is implicated some conserved law, the other side every conserved law is based on one of symmetry of nature"

(A) symmetry under transformation:

$$[\hat{Q}, \hat{H}] = 0 \quad (57)$$

(B) symmetry with conserved observable:

$$[\hat{F}, \hat{H}] = 0 \quad (58)$$

(C) symmetry of space translation:

$$[p, \hat{H}] = 0 \quad (59)$$

equaly with Momentum Conserved.

(D) symmetry of space turning translation:

$$[\vec{l}, \hat{H}] = 0 \quad (60)$$

equaly to angular momentum conserved.

(E) Some other symmetry in QM:

5.5 *Commutation

Here is an example.

- (5) 时间均匀性与能量守恒
- (6) 空间反射对称性与宇称守恒
- (7) 同位旋空间旋转对称性与同位旋守恒

Figure 1: picture from teacher's ppt

6 Identical Particle System and The Wavefunction

6.1 identical particle and exchange operator

An identical particle is the particle that have all the same intrinsic properties with other particle. Two identical particles are indistinguishable in QM. Define an exchange operator:

$$P_{ij}\psi(q_1 \dots q_i \dots q_j \dots) = \psi(q_1 \dots q_j \dots q_i \dots) \quad (61)$$

For any observable in a system with identical particles, the observables have symmetry of exchange

$$P_{ij}F(q_1 \dots q_i \dots q_j \dots) = F(q_1 \dots q_i \dots q_j \dots) \quad (62)$$

$$[p_{ij}, H] = 0 \quad (63)$$

For a system of identical particles, we set if an exchange of identical particles happens, then the wavefunction of this system will only differ by times a constant.

$$P_{ij}\psi = C\psi \quad (64)$$

$$P_{ij}P_{ij}\psi = C^2\psi = \psi$$

$$C = \pm 1 \quad (65)$$

So that means only two possibilities will happen: the wavefunction is symmetric: $\psi = P_{ij}\psi$, or the wavefunction is anti-symmetric: $-\psi = P_{ij}\psi$.

6.2 properties of Boson and Fermion

Boson is the particle with spin is integer times of $\hbar$. And Fermion is the particle that the spin is half-integer times of $\hbar$. But a very important nature of Boson and Fermion is Identical Boson's exchange always symmetry for a system with Boson identical particles, and identical Fermion's exchange in a system with identical Fermion always anti-symmetric.

For Boson:

$$\begin{aligned} &\text{if } k_1 \neq k_2 \\ \psi_{k_1 k_2}^S(q_1, q_2) &= \frac{1}{\sqrt{2}}(1 + P_{12})\phi_{k_1}(q_1)\phi_{k_2}(q_2) \end{aligned} \quad (66)$$

$$\begin{aligned} &\text{if } k_1 = k_2 \\ \psi_{k_1 k_2}^S(q_1, q_2) &= \phi_{k_1}(q_1)\phi_{k_2}(q_2) \end{aligned} \quad (67)$$

For Fermion:

$$\psi_{k_1 k_2}^A(q_1, q_2) = \frac{1}{\sqrt{2}}(1 - P_{12})\phi_{k_1}(q_1)\phi_{k_2}(q_2) \quad (68)$$

and we can see the state of Fermion at $k_1 = k_2$ is not exist. **This is the Pauli principle.**

7 Matrix

7.1 The image

Just like tensor , the image can be analog as the component, we can write the operator as the :

$$A = \hat{A}_b^a \quad ; \text{ and the component: } \hat{A}_j^i = \hat{A}_b^a \phi_a^{*i} \phi_j^b \quad (69)$$

The ϕ here can be seen as the vector in GR, and the ϕ^* can be seen as the dual vector in GR. So we changed all old ways to calculate the QM questions.

7.2 relationship under a image

7.2.1 Image theory

Now we can write clearly the relationship of these operator under special image , but at first we must supplemt a proposition:

If ϕ is the eigenstate of an operator then the operator will be a diagonal matrix:

$$\phi^{*i} \hat{F} \phi_j = \hat{F}_j^i = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n) \quad (70)$$

7.2.2 Eigenstate equation under a not eigenstate image

The eigenstate equation is $\hat{F}\phi = \lambda\phi$, and under the matrix way to express , we can write it as:

$$\hat{F}_b^a \phi_j^b = \lambda_J \phi_j^a \quad (71)$$

and now we set the ψ as the base of image the we get the decompositon equation:

$$\hat{F}_b^a \phi_j^b = \lambda_J \phi_j^a = \hat{F}_k^i \psi_b^{*k} \psi_i^a \phi_j^b = \lambda_J \phi_j^a \quad (72)$$

and the eigenstate can be decomposition in new base too:

$$\phi_i^a = \phi_i^k \psi_k^a \quad (73)$$

consider all of them ,we get the final equation:

$$\hat{F}_k^i \psi_i^a \phi_j^k = \lambda_J \phi_j^a = \lambda_J \phi_j^k \psi_k^a = \lambda_J \phi_j^k \delta_k^i \psi_i^a \quad (74)$$

$$\hat{F}_k^i \phi_j^k \psi_i^a - \lambda_J \phi_j^k \delta_k^i \psi_i^a = 0 \quad (75)$$

$$(\hat{F}_k^i - \lambda_J \delta_k^i) \phi_j^k \psi_i^a = 0 \quad (76)$$

and if eq(76) have the non-normal solve of ϕ_j^k ,them the determinant of matrix $(\hat{F}_k^i - \lambda_J \delta_k^i)$ is equal to 0:

$$\det(\hat{F}_j^i - \lambda_J \delta_j^i) = 0 \quad (77)$$

Which can solve the eigenstate and the eigenvalue out from base of $\hat{F}_j^i$.

7.2.3 *An example of calculation

$\psi = c_1 Y_{1-1} + c_2 Y_{11}$ express the probabily value of l_x

$$\psi = c_1 Y_{1-1} + c_2 Y_{11} = c_1 \psi_1 + c_2 \psi_3$$

$$\overline{l_x} = \psi^* l_x \psi = \psi^* \frac{1}{2} (l_+ + l_-) \psi$$

$$(l_x)_{11} = \frac{1}{2} \psi^{*1} (l_+ + l_-) \psi_1 = 0; \quad (l_x)_{12} = \frac{1}{2} \psi^{*1} (l_+ + l_-) \psi_2 = \frac{1}{2} \sqrt{1(2) + 0} \hbar$$

$$= \frac{\sqrt{2}}{2} \hbar; \quad (l_x)_{13} = \psi^{*1} l_x \psi_3 = \frac{1}{2} \psi^{*1} (l_+ + l_-) \psi_3 = 0$$

$$(l_x)_{22} = 0; \quad (l_x)_{23} = \frac{1}{2} \psi^{*2} (l_+ + l_-) \psi_3 = \frac{1}{2} \sqrt{2+0} \hbar = \frac{\sqrt{2}}{2} \hbar; \quad (l_x)_{33} = \frac{1}{2} \psi^{*3} (l_+ + l_-) \psi_3 = 0$$

$$l_{x,ij} = \begin{bmatrix} 0 & \frac{\sqrt{2}}{2} \hbar & 0 \\ \frac{\sqrt{2}}{2} \hbar & 0 & \frac{\sqrt{2}}{2} \hbar \\ 0 & \frac{\sqrt{2}}{2} \hbar & 0 \end{bmatrix}; \quad \begin{vmatrix} -\lambda & \frac{\sqrt{2}}{2} \hbar & 0 \\ \frac{\sqrt{2}}{2} \hbar & -\lambda & \frac{\sqrt{2}}{2} \hbar \\ 0 & \frac{\sqrt{2}}{2} \hbar & -\lambda \end{vmatrix} = 0; \quad -\lambda(\lambda^2 - \frac{1}{2} \hbar^2) + \frac{1}{2} \lambda \hbar^2 = 0;$$

$$\lambda_0 = 0; \quad \lambda_1 = \hbar; \quad \lambda_{-1} = -\hbar;$$

$$\begin{bmatrix} 0 & \frac{\sqrt{2}}{2} \hbar & 0 \\ \frac{\sqrt{2}}{2} \hbar & 0 & \frac{\sqrt{2}}{2} \hbar \\ 0 & \frac{\sqrt{2}}{2} \hbar & 0 \end{bmatrix} \vec{v}_0 = 0; \quad v_0^2 = 0, \quad v_0^1 = -v_0^3, \quad \vec{v}_0 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ -\frac{1}{\sqrt{2}} \end{bmatrix}$$

$$\begin{bmatrix} -\hbar & \frac{\sqrt{2}}{2} \hbar & 0 \\ \frac{\sqrt{2}}{2} \hbar & -\hbar & \frac{\sqrt{2}}{2} \hbar \\ 0 & \frac{\sqrt{2}}{2} \hbar & -\hbar \end{bmatrix} \vec{v}_1 = 0; \quad v_1^1 = \frac{\sqrt{2}}{2} v_1^2; \quad v_1^3 = \frac{\sqrt{2}}{2} v_1^2; \quad \vec{v}_1 = \begin{bmatrix} \frac{1}{2} \\ \frac{\sqrt{2}}{2} \\ \frac{1}{2} \end{bmatrix}$$

$$\begin{bmatrix} \hbar & \frac{\sqrt{2}}{2} \hbar & 0 \\ \frac{\sqrt{2}}{2} \hbar & \hbar & \frac{\sqrt{2}}{2} \hbar \\ 0 & \frac{\sqrt{2}}{2} \hbar & \hbar \end{bmatrix} \vec{v}_{-1} = 0; \quad -v_{-1}^1 = \frac{\sqrt{2}}{2} v_{-1}^2; \quad -v_{-1}^3 = \frac{\sqrt{2}}{2} v_{-1}^2; \quad \vec{v}_{-1} = \begin{bmatrix} \frac{1}{2} \\ -\frac{\sqrt{2}}{2} \\ \frac{1}{2} \end{bmatrix}$$

7.2.4 transformation between image

Just like tensor transformation :

$$\psi_i^a = \psi_i^k \phi_k^a; \psi_i^k = \psi_i^a \phi_a^{*k} \quad (78)$$

and one more thing, a special symbol is also very similar with GR:

$$\psi_a^{*k} \psi_k^b = \delta_a^b \quad (79)$$

and ψ is a set of base of any image.

8 Spin and angular momentum

Spin can be seen as the new dimension of freedom. The spin operator $\vec{S}$ and for any direct, the projection can only be one of two values: $\frac{1}{2}\hbar$ and $-\frac{1}{2}\hbar$:

$$S_x = \pm \frac{1}{2}\hbar; S_y = \pm \frac{1}{2}\hbar; S_z = \pm \frac{1}{2}\hbar; S_n = \pm \frac{1}{2}\hbar \quad (80)$$

and the spin magnetic moment :

$$M_s = -\frac{e}{\mu} \vec{S} \quad (81)$$

8.1 the wavefunction of spin

There are two ways to write the wavefunction of spin one of them is by written into a row:

$$\psi = a\chi_{\pm\frac{\hbar}{2}} + b\chi_{-\frac{\hbar}{2}} \quad (82)$$

and the 'a' and 'b' here is the coefficient and the χ is eigenstate of operator S_z . That :

$$S_z \chi_{\pm\frac{\hbar}{2}} = m_s \hbar \chi_{\pm\frac{\hbar}{2}} = \pm \frac{\hbar}{2} \chi_{\pm\frac{\hbar}{2}} \quad ; m_s = \pm \frac{1}{2} \quad (83)$$

so we can write it into simple form:

$$S_z \chi_{m_s} = m_s \hbar \chi_{m_s} ; (-s < m_s < s) \quad (84)$$

$$S^2 \chi_{m_s} = s(s+1) \hbar^2 \chi_{m_s} ; s = \frac{1}{2} \quad (85)$$

and similar with the $\vec{l}$, $\vec{S}$ also have the relationship:

$$[S_i, S_j] = i\hbar \varepsilon_{ijk} S_k;$$

$$S_+ = S_x + iS_y; S_- = S_x - iS_y$$

and the commutator relationship:

$$[S^2, S_z] = 0; [l_i, S_j] = 0 \quad (86)$$

8.2 Pauli image

$$\sigma = \frac{2}{\hbar} s \quad (87)$$

and Pauli image in fact just the dimensionless form of s.

$$\sigma_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \sigma_y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \quad (88)$$

one more important thing

$$\sigma_+ = \frac{1}{2}(\sigma_x + i\sigma_y); \sigma_- = \frac{1}{2}(\sigma_x - i\sigma_y) \quad (89)$$

so the different between s and σ is the dimension:

$$\sigma_x = \sigma_+ + \sigma_-; \sigma_y = \frac{1}{i}(\sigma_+ - \sigma_-) \quad (90)$$

$$s_x = \frac{1}{2}(s_+ + s_-); s_y = \frac{1}{2i}(s_+ - s_-) \quad (91)$$

the coefficient is quite different here. and what's more:

$$\sigma_+ \chi_- = \chi_+; \sigma_- \chi_+ = \chi_- \quad (92)$$

$$s_+ \chi_- = \hbar \chi_+; s_- \chi_+ = \hbar \chi_- \quad (93)$$

8.3 angular momentum

Then we can define the angular momentum:

$$\vec{j} = \vec{s} + \vec{l} \quad (94)$$

$$(95)$$

a very important thing the co-eigenstate of j^2 and j_z is :

$$\phi_{lmj} = C_1 Y_{lm} \chi_{\frac{1}{2}} + C_2 Y_{lm+1} \chi_{-\frac{1}{2}} \quad (96)$$

$$j_z Y_{lm+1} \chi_{-\frac{1}{2}} = (m + \frac{1}{2}) \hbar Y_{lm+1} \chi_{-\frac{1}{2}} \quad (97)$$

$$j_z Y_{lm} \chi_{\frac{1}{2}} = (m + \frac{1}{2}) \hbar Y_{lm} \chi_{\frac{1}{2}} \quad (98)$$

notice that the wavefunction $Y_{lm} \chi_{\frac{1}{2}}$ and $Y_{lm+1} \chi_{-\frac{1}{2}}$ have the same eigenvalue of j_z and the co-eigenstate of both (L^2, j^2, j_z) is $\phi = c_1 Y_{lm} \chi_{\frac{1}{2}} + c_2 Y_{lm+1} \chi_{-\frac{1}{2}}$ which can write into short form as : ϕ_{lmj} and we can also get the simple conclusion and list all difference between three type of angular momentum:

$\hat{L}$	$\hat{S}$	$\hat{J}$
l	$s = \frac{1}{2}$	$j = l + \frac{1}{2}/j = l - \frac{1}{2}$
m	m_s	m_j
$-l, \dots, l$	$-\frac{1}{2}, \frac{1}{2}$	$-j, \dots, j$
$[L^2, L_i] = 0$	$[S^2, S_i] = 0$	$[J^2, J] = 0$
$[L_i, L_j] = i\hbar\epsilon_{ijk}L_k$	$[S_i, S_j] = i\hbar\epsilon_{ijk}S_k$	$[J_i, J_j] = i\hbar\epsilon_{ijk}J_k$
(L^2, L_z)	(S^2, S_z)	(J^2, J_z)
Y_{lm}	$\chi_{\frac{1}{2}}, \chi_{-\frac{1}{2}}$	ψ_{ljm_j}
$L^2 Y_{lm} = l(l+1)\hbar^2 Y_{lm}$	$S^2 \chi_{m_s} = s(s+1)\hbar^2 \chi_{m_s}$	$J^2 \psi_{ljm_j} = j(j+1)\hbar^2 \psi_{ljm_j}$
$L_z Y_{lm} = m\hbar Y_{lm}$	$S_z \chi_{m_s} = m_s \hbar \chi_{m_s}$	$J_z \psi_{ljm_j} = m_j \hbar \psi_{ljm_j}$
		$L^2 \psi_{ljm_j} = l(l+1)\hbar^2 \psi_{ljm_j}$

9 One Dimensional Particle in Potential Field

9.1 Infinity Potential Well

The V satisfied the define below is called the one dimensional infinity potential well:

Type one: The well is in one site and it's not symmetry with the coordinate frame.

$$V = \begin{cases} 0 & 0 < x < a. \\ \infty & a < x. \end{cases} \quad (99)$$

Which means the Shrödinger equation will be :

$$-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \psi + (V - E)\psi = 0 \quad (100)$$

Consider the conditon :

I: The wavefunction must be zero at $x = 0$ and $x = a$.

II: The wavefunction must be the eigenstate of energy.

The solution:

$$\psi = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi}{a}x\right); \frac{n\pi}{a} = \sqrt{\frac{2mE}{\hbar^2}} \quad (101)$$

Type two : The well is symmetry at both side of x line.

$$V = \begin{cases} 0 & |x| > a; \\ \infty & |x| < a. \end{cases} \quad (102)$$

The Shrödinger equation is as same as eq.100, but the boarder condition is not the same so the result is not the same.

The solution:

$$\psi = \begin{cases} \sqrt{\frac{2}{a}} \sin(\frac{k\pi}{2a}x), k = 0, 2, 4, \dots \\ \sqrt{\frac{2}{a}} \cos(\frac{k\pi}{2a}x), k = 1, 3, 5, \dots \end{cases} \quad (103)$$

Reason of why the wavefunction must be $\cos x$ or $\sin x$ but can not be like $A\sin x + B\cos x$ is the **wavefunction must have the precision parity if the potential function is symmetry in the space**

9.2 Limited depth potential well

The potential well called limited depth if V satisfied :

$$V = \begin{cases} V & |x| > \frac{a}{2} \\ 0 & |x| < \frac{a}{2} \end{cases} \quad (104)$$

and V here is a real number and not divergent.

In this situation the Shrödinger equation can be written as:

$$-\frac{\hbar^2}{2m} \nabla^2 \psi + (V - E)\psi = 0 \quad (105)$$

and now I will write down the process of solve:

Solve:

$$\begin{cases} \nabla^2 \psi + \frac{2m}{\hbar^2} E \psi = 0 & |x| < \frac{a}{2} \\ \nabla^2 \psi + \frac{2m}{\hbar^2} (E - V) \psi = 0 & |x| > \frac{a}{2} \end{cases}$$

and the area $|x| < \frac{a}{2}$ still be $\sin$ or $\cos$ and still can not be both:

$$\psi = \begin{cases} B \sin \sqrt{\frac{2mE}{\hbar^2}} x & |x| < \frac{a}{2} \\ B \cos \sqrt{\frac{2mE}{\hbar^2}} x & |x| < \frac{a}{2} \end{cases}$$

at the left half:

$$\psi = A e^{\sqrt{\frac{2m(E-V)}{\hbar^2}} x} \quad |x| < -\frac{a}{2}$$

at the right half :

$$\psi = C e^{-\sqrt{\frac{2m(E-V)}{\hbar^2}} x} \quad |x| > \frac{a}{2}$$

and we can solve the wavefunction by steps below:

I: Choose a parity.

II: use the bound condition :

A:

$$\begin{cases} Ae^{-K\frac{a}{2}} = B\sin/\cos(-K\frac{a}{2}) \\ B\sin/\cos(K\frac{a}{2}) = Ce^{-K\frac{a}{2}} \end{cases}$$

B: The derivative of wavefunction is continuous

The point needed to be noticed is, the parity of wavefunction under a symmetry potential.

9.3 Reflection and Transmission of the Square Potential Well

Now consider the square potential like below:

$$V(x) = \begin{cases} V_0 & 0 < x < a \\ 0 & x < 0, \quad x > a \end{cases} \quad (106)$$

and we can write the wavefunction as three parts: the transmission part, the reflection part and the origin part:

$$\psi(x) = \begin{cases} e^{ipx/\hbar} + Re^{-ipx/\hbar} \\ Te^{-ipx/\hbar} \end{cases} \quad (107)$$

the coefficient defined like:

$$\text{Transmission coefficient} := |T|^2 \quad (108)$$

$$\text{Reflection coefficient} := |R|^2 \quad (109)$$

And the part through the potential is:

$$\psi = Ae^{i\sqrt{\frac{2m(E-V)}{\hbar^2}}x} + Be^{-i\sqrt{\frac{2m(E-V)}{\hbar^2}}x} \quad (110)$$

Notice: this solution is based on the solve of Hamilton equation so the momentum of every part is the same.

And we can solve it by the bound condition and continuous condition with 4 equations we can solve whole 4 coefficients.

9.4 The Delta Potential Wall

The potential function is satisfied below call the "Delta Potential Wall"

$$V(x) = \gamma\delta(x) \quad \gamma > 0 \quad (111)$$

And noticed that the continuous of wavefunction is special that: the two class derivative of the wavefunction is not existed at the $x = 0$ point. So use the new continuous condition below to replace the two class derivative condition:

$$\psi'(0^+) - \psi'(0^-) = \frac{2m\gamma}{\hbar^2}\psi(0) \quad (112)$$

Proof. How to get this equation:

$$\begin{aligned} \int_{-\epsilon}^{\epsilon} \left[\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} (E - \gamma\delta(x)) \right] \psi &= 0 \\ \Rightarrow \psi'(0^+) - \psi'(0^-) - \frac{2m}{\hbar^2} \gamma \psi &= 0 \end{aligned}$$

□

For this situation also considering the reflection and transmission of wavefunction:

$$\psi = \begin{cases} e^{ipx/\hbar} + Re^{-ipx/\hbar} & x < 0 \\ Te^{ipx/\hbar} & x > 0 \end{cases} \quad (113)$$

And the wavefunction can be solved by:

$$\begin{cases} 1 + R = T \\ \psi(0) = T \\ ip/\hbar T - (ip/\hbar - Rip/\hbar) = \frac{2m\gamma}{\hbar^2} \psi(0) \end{cases} \quad (114)$$

9.4.1 The Bound State of Delta Potential Well

Now we consider the Potential as :

$$V = -\gamma\delta(x) \quad \gamma > 0 \quad (115)$$

Here is the solution:

$$\psi(x) = \begin{cases} Ae^{i\sqrt{\frac{2mE}{\hbar^2}}x} & x < 0 \\ Ae^{-i\sqrt{\frac{2mE}{\hbar^2}}x} & x > 0 \end{cases} \quad (116)$$

This is a bound state ,althought the potential function is symmetry with the x-axis but the odd parity can not be exist because the coefficient will be zero.

9.5 The Leap Condition

The condition of transformation from square to delta potential.

$$\lim_{\epsilon \rightarrow 0^+} [\psi'(\epsilon) - \psi'(-\epsilon)] = \kappa(A + B)[(1 + \kappa\epsilon) - (1 - \kappa\epsilon)] = \frac{2m\gamma}{\hbar^2} \psi(0) \quad (117)$$

10 One Dimension Harmonic Oscillator

The V satisfied below:

$$V = \frac{1}{2}Kx^2 = \frac{1}{2}m\omega^2x^2 \quad \omega = \sqrt{\frac{K}{m}} \quad (118)$$

So the Hamilton equation is :

$$-\frac{\hbar^2}{2m}\nabla^2\psi + \frac{1}{2}m\omega^2x^2\psi = E\psi \quad (119)$$

The solve of this is complexed and we will show the result:

$$\alpha = \sqrt{\frac{m\omega}{\hbar}} \quad \xi = \alpha x \quad (120)$$

$$\text{The equation become: } \frac{d^2\psi(x)}{d\xi^2} + [\lambda - \xi^2]\psi(x) = 0 \quad (121)$$

$$\psi = \left(\frac{\alpha}{\sqrt{\pi}2^n n!}\right)^{\frac{1}{2}} e^{-\frac{1}{2}\alpha^2 x^2 - \frac{i}{\hbar}E_n t} H_n(\alpha x) \quad (122)$$

Here will write some train of thought to get the solutino:

Consider the $x \rightarrow \infty$ and the equation become:

$$\frac{d^2\psi}{d\xi^2} = \xi^2\psi \quad (123)$$

and get the result under the ∞ condition:

$$\psi_{\infty} = c_1 e^{-\xi^2/2} + c_2 e^{\xi^2/2} \quad (124)$$

consider that if the $\xi \rightarrow \infty$ and the wavefunction still be convergent so $c_2 = 0$.

$$\psi = c_1 e^{-\xi^2/2} \quad (125)$$

Now the precision resolution is become:

$$\psi = H(\xi)e^{-\xi^2/2} \quad (126)$$

10.1 Some Hermit Polynomial

$$H_0 = 1 \quad (127)$$

$$H_1 = 2\xi \quad (128)$$

$$H_2 = 4\xi^2 - 2 \quad (129)$$

10.2 Energy of Harmonic Oscillator

$$\text{consider: } \lambda = \frac{2E}{\omega\hbar}; \lambda_n = 2n + 1 \quad (130)$$

$$E_n = \left(n + \frac{1}{2}\right)\hbar\omega \quad n = 0, 1, 2 \dots \quad (131)$$

Notice that the define of n is from the solve of Hermit equations, and this make the energy become the separation value

11 The Algebraic Representation of Hamiltonian

The Hamiltonian of one dimensional Harmonic Oscillator can be present as :

$$H = \frac{1}{2\mu}p^2 + \frac{1}{2}\mu\omega^2x^2 \quad (132)$$

Define the brith operator and annihilation operator under the **nature units**:

$$a = \frac{1}{\sqrt{2}}(x + ip) \quad a^\dagger = \frac{1}{\sqrt{2}}(x - ip) \quad (133)$$

and back to the standard units:

$$x = \frac{1}{\sqrt{2}}\sqrt{\frac{\hbar}{\omega\mu}}(a + a^\dagger) \quad (134)$$

$$p = \frac{1}{\sqrt{2}i}\sqrt{\hbar\omega\mu}(a - a^\dagger) \quad (135)$$

and thes two operator is very inportant , the a called the brith operator of particles, and $a^\dagger$ called the annihilation operator of particles.Before explain the reason , I will list some relation between a and $a^\dagger$:

$$[a, a^\dagger] = 1 \quad (136)$$

Proof.

$$\begin{aligned} [a^\dagger, a] &= \left[\frac{1}{\sqrt{2}}(x - ip), \frac{1}{\sqrt{2}}(x + ip) \right] = \frac{1}{2}([(x - ip), (x + ip)]) \\ &\Rightarrow \frac{1}{2}([x, ip] - [ip, x]) = [x, ip] = i\hbar \times i = -1 \end{aligned}$$

□

And consider the Hamiltonian eq.132 we replace with eq.134 and eq.135:

$$H = (a^\dagger a + \frac{1}{2}) = \hat{N} + \frac{1}{2} \quad (137)$$

and the $\hat{N} = a^\dagger a$ the reason of the Hamiltonian is the eigenvalue of harmonic oscillator is $n + \frac{1}{2}$ under hte nature units.

11.1 The Occupation Number representation

If the wavefunction satisfied :

$$\hat{N}\psi = n\psi \quad (138)$$

then write the wavefunction as the $|n\rangle$ and called the Occupation Number Representation wavefunction. And we can know that

$$a^\dagger |n\rangle \sim |n+1\rangle \quad (139)$$

$$a |n\rangle \sim |n-1\rangle \quad (140)$$

Proof.

$$\begin{aligned} [\hat{N}, a] = -a &\Rightarrow [\hat{N}, a]|n\rangle = -a|n\rangle \\ &= (\hat{N}a - a\hat{N})|n\rangle = \hat{N}a|n\rangle - na|n\rangle = -a|n\rangle \\ \hat{N}a|n\rangle &= (n-1)a|n\rangle \\ a|n\rangle &= |n-1\rangle \end{aligned}$$

and for $a^\dagger$ is the same . □

And we know the coefficient of $a|n\rangle$:

$$|n\rangle = \frac{1}{\sqrt{n!}} a^{\dagger n} |0\rangle \quad (141)$$

$$a|n\rangle = \sqrt{n} |n-1\rangle \quad (142)$$

$$a^\dagger |n\rangle = \sqrt{n+1} |n+1\rangle \quad (143)$$

Proof.

$$|1\rangle = C a^\dagger |0\rangle$$

$$1 = C^2 \langle a^\dagger |0\rangle \langle a^\dagger |0\rangle = C^2 \langle 0|0\rangle = 1 \Rightarrow C = 1$$

$$1 = C^2 \langle a^{\dagger 2} |0\rangle \langle a^{\dagger 2} |0\rangle = C^2 \langle (\hat{N}+1)a^\dagger |0\rangle \langle a^\dagger |0\rangle \Rightarrow 2C^2 = 1 \Rightarrow C = \frac{1}{\sqrt{2}}$$

$$\text{set for any } n: 1 = \langle a^{\dagger n} |0\rangle \langle a^{\dagger n} |0\rangle$$

$$\begin{aligned} 1 &= \langle n|n\rangle = \langle a^{\dagger n} |0\rangle \langle a^{\dagger n} |0\rangle = C^2 \langle a a^{\dagger} a^{\dagger n-1} |0\rangle \langle a^{\dagger n-1} |0\rangle = C^2 \langle (\hat{N}+1)a^{\dagger n-1} |0\rangle \langle a^{\dagger n-1} |0\rangle \\ &= C^2 n - 1! + C^2 (n-1)(n-1!) = 1 = n! C^2 \Rightarrow C = \frac{1}{\sqrt{n!}} \end{aligned}$$

$$1 = n! \langle a^{\dagger n} |0\rangle \langle a^{\dagger n} |0\rangle$$

$$|n\rangle = \frac{1}{\sqrt{n!}} a^{\dagger n} |0\rangle$$

and for eq.142 and 143:

$$a^\dagger |n\rangle = a^\dagger \frac{1}{\sqrt{n!}} a^{\dagger n} |0\rangle = \frac{1}{\sqrt{n!}} a^{\dagger n+1} |0\rangle = \frac{1}{\sqrt{n!}} \sqrt{n+1}! |n+1\rangle = \sqrt{n+1} |n+1\rangle$$

and eq 142 is as the similar way as the eq 143. □

11.2 The Commutator Relationship Between Normal Angular Momentum Operators

$$[j_i, j_j] = i\hbar\epsilon_{ijk}j_k \quad i, j, k = 1, 2, 3 \quad (144)$$

$$j^2 = j_x^2 + j_y^2 + j_z^2 \quad (145)$$

$$[j^2, j_i] = 0 \quad i = 1, 2, 3 \quad (146)$$

define:

$$j_{\pm} = j_x \pm ij_y \quad (147)$$

and :

$$[j_z, j_{\pm}] = \pm\hbar j_{\pm} \quad (148)$$

$$j_{\pm}j_{\mp} = j^2 - j_z^2 \pm \hbar j_z \quad (149)$$

12 Two Dimensional Phonon System

Operators of two phonons will not effect on each other so we mark them as:

$$a_1, a_1^{\dagger} \quad \hat{N}_1 = a_1^{\dagger}a_1 \quad (150)$$

$$a_2, a_2^{\dagger} \quad \hat{N}_2 = a_2^{\dagger}a_2 \quad (151)$$

and the eigenvalue of operator $\hat{N}$ is n_1 and n_2 which is a positived integer. and as the same we define the $|n\rangle$ as :

$$|n_1 n_2\rangle = \frac{a_1^{\dagger n_1} a_2^{\dagger n_2}}{\sqrt{n_1! n_2!}} |00\rangle \quad (152)$$

and we define the "angular momentum" of two particla system:

$$j_x = \frac{1}{2}(a_1^{\dagger}a_2 + a_2^{\dagger}a_1) \quad (153)$$

$$j_y = \frac{1}{2i}(a_1^{\dagger}a_2 - a_2^{\dagger}a_1) \quad (154)$$

$$j_z = \frac{1}{2}(a_1^{\dagger}a_1 - a_2^{\dagger}a_2) = \frac{1}{2}\hat{N}_1 - \hat{N}_2 \quad (155)$$

$$j_+ = a_1^{\dagger}a_2 \quad j_- = a_2^{\dagger}a_1 \quad (156)$$

and this kind of define is legal which is satisfied all equations that angular momentum should follow.

Consider the eigenvalue of j^2 , we can define that :

$$j = \frac{n}{2} = \frac{n_1 + n_2}{2} = \begin{cases} 0, 1, 2, 3 \dots \\ \frac{1}{2}, \frac{3}{2}, \frac{5}{2} \dots \end{cases} \quad (157)$$

And because that we can replace the n into j and m to express the wavefunction:

$$j = \frac{1}{2}(n_1 + n_2), m = \frac{1}{2}(n_1 - n_2) \quad (158)$$

so the wavefunction can be written as :

$$|jm\rangle = \frac{(a^\dagger)^{j+m}(a^\dagger)^{j-m}}{\sqrt{(j+m)!(j-m)!}}|0\rangle \quad (159)$$

$$j_+|jm\rangle = a_1^\dagger a_2|jm\rangle = \sqrt{j(j+1) - m(m+1)}\hbar|jm+1\rangle \quad (160)$$

$$j_-|jm\rangle = a_1 a_2^\dagger|jm\rangle = \sqrt{j(j+1) - m(m-1)}\hbar|jm-1\rangle \quad (161)$$